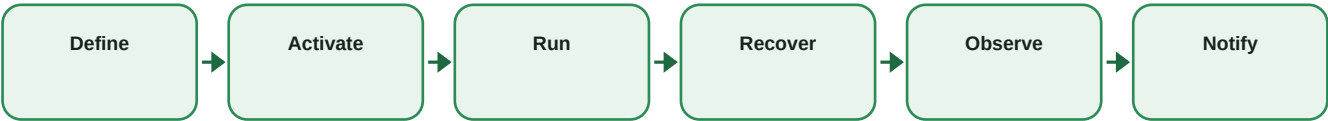


PacketAgent Canonical Worker Implementation Report

A concise engineering and product account of W1 through W10.2, the safety model they establish, the evidence behind the pause, and the exact work that remains.

W1-W8 complete	W9 complete	W10.1-W10.2 complete
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Pause state	Implementation is committed through W10.2. The next unchecked backlog item is W10.3 PacketPhone controls. The unified Packet-suite lifecycle is not yet a shipped claim.
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Prepared from committed repository state b259a4f on 2026-07-28.

Executive snapshot

PacketAgent now has a canonical, durable, bounded Worker runtime and a locally certified PacketADE-to-PacketAgent handoff. W10.2 adds the PacketAgent side of PacketChat delivery. PacketPhone controls and live cross-product certification remain.

Area	State	What is true at the pause
Canonical Worker	Complete	Versioned identity, policy, lifecycle, activation, run, and checkpoint records.
Durable runtime	Complete	Bounded supervisor, restart recovery, effect receipts, rolling budgets, and fenced leases.
Operator safety	Complete	Attention, approvals, pause/resume/stop/revoke, immutable audit, and race coverage.
Answerability	Complete	Evidence journal, deterministic rollups, retention, read model, SSE, and Worker UI.
PacketADE	Complete locally	Strict WorkerPackage v1 receiver, trust boundary, deployment API, reconnectable events, restart handoff gate.
PacketChat	Complete in PacketAgent	Encrypted route configuration, bounded cards, hardened delivery, authenticated read-only callbacks.
PacketPhone	Not implemented	W10.3 will add replay-proof mutating controls through W7.
Cross-product live gate	Not complete	W10.4 needs real PacketChat and PacketPhone endpoints plus credentials.

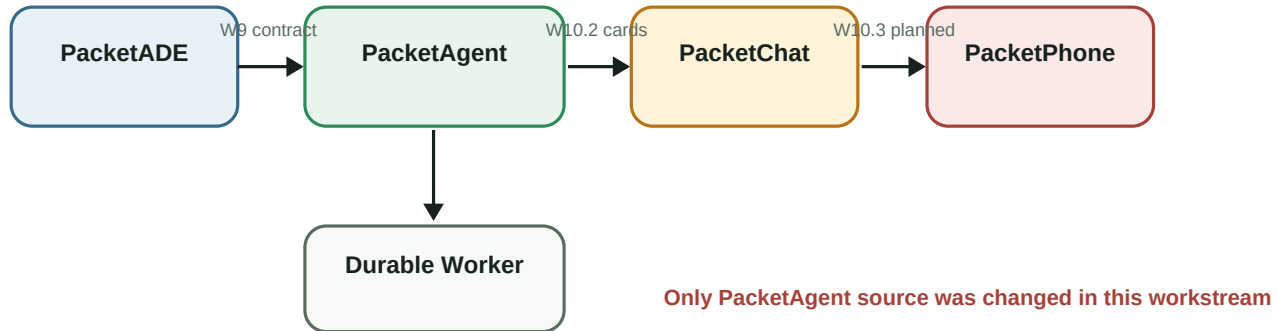
Quality gate

1,493 API tests passed, 2 intentionally skipped, 0 failed. All 28 web tests passed. Typecheck and production web build passed. Lint has 0 errors and 145 inherited warnings.

The implementation pause is deliberate: BACKLOG.md is now the single remaining-work ledger, and dev/CODEX-HANDOFF.md points directly to its W10.3 entry.

Product model and non-negotiable invariants

PacketAgent is the always-available control plane. Workers are event-driven and finite, not endless token loops.



Every autonomous Worker is required to be:

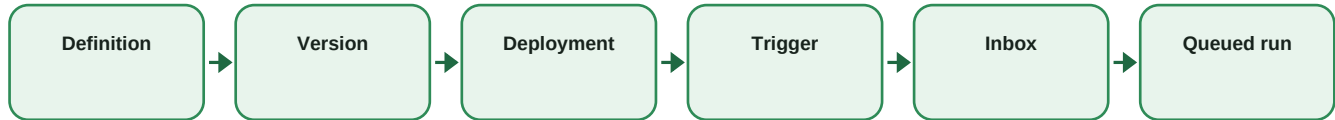
- **Bounded** by elapsed time, iterations, provider cost, consecutive failures, tool calls, retry attempts, and permissions.
- **Permissioned** immediately before each tool or external action at the runtime boundary.
- **Resumable** from a durable, immutable checkpoint with a fenced lease.
- **Duplicate-safe** through effect receipts and stable idempotency identities.
- **Auditable** through monotonic events, evidence, provenance, rollups, and redacted read models.
- **Stoppable** independently of the authoring client through pause, stop, and deployment revocation.

Naming rule

All new public code and persisted identifiers use PacketAgent, packetagent, and PACKETAGENT_*. The old TaskLoom name survives only for migration compatibility and historical records.

W1-W3: identity, persistence, and activation

These loops turn an authored definition into one durable, version-pinned queued run.



W1 - Canonical contract

- Introduced versioned WorkerDefinition, WorkerVersion, WorkerDeployment, WorkerTrigger, WorkerPolicy, WorkerRun, and WorkerCheckpoint schemas.
- Made deployed versions immutable and encoded legal lifecycle and terminal transitions.
- Preserved provenance from PacketADE and legacy read projections without creating new legacy APIs.

W2 - Persistence and lifecycle

- Added one storage-neutral repository contract over JSON, SQLite, and managed Postgres.
- Implemented idempotent lifecycle commands, command receipts, rollback links, private RBAC routes, and backend parity.
- Added optimistic revision conflicts and multi-writer activation protection.

W3 - Activation intake

- Created a versioned activation envelope and deduplicated delivery inbox.
- Admitted manual, cron, webhook, queue, and alert deliveries through one atomic path.
- Encrypted sensitive activation payloads behind expiring references and created run plus execution job atomically.

W4-W5: bounded execution and crash safety

The runtime can make progress, stop at every boundary, and recover without silently abandoning work or repeating an uncertain mutation.

W4 - Bounded supervisor

- Isolated provider, tool, checkpoint, lease, clock, and cancellation ports.
- Implemented plan -> act -> evaluate -> checkpoint -> decide phases with explicit finite budgets.
- Added optimistic run revisions, renewable fencing leases, scheduler cancellation, and forced termination of ignored aborts.
- Closed adversarial cases for endless tools, hung providers, invalid evaluations, exact cost exhaustion, cancellation at every phase, and lease theft.

W5 - Recovery and external effects

- Persisted digest-chained full-state checkpoints with exact phase cursor, memory, completed actions, approvals, artifacts, effects, and remaining budget.
- Wrapped mutating tools in prepared/completed effect receipts with stable action identities.
- Recovered expired work from the exact cursor, reconciled provable completions, retried only safe idempotent effects, and quarantined uncertain replay.
- Added startup and periodic recovery plus cross-backend crash injection coverage.

Operational result

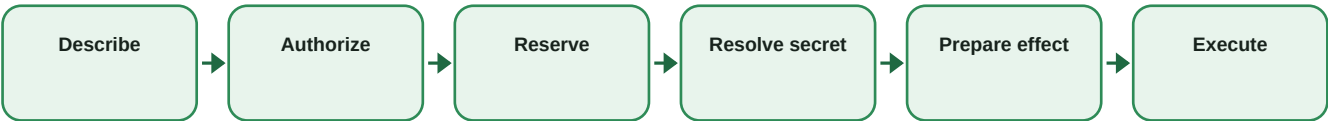
Closing PacketADE, PacketChat, the browser, or the originating authoring app does not define Worker liveness. Durable state and the PacketAgent scheduler do.

W6: permission, secret, network, process, and budget enforcement

W6 moved safety from descriptive configuration to mandatory runtime boundaries.

Slice	Enforcement delivered
W6.1	Compiled version-digest-bound tool, verb, resource, effect, and approval tuples. Deployment grants can narrow but never broaden.
W6.2	Normalized each operation and re-authorized it immediately before the registered handler. Denials are redacted Worker events.
W6.3	Resolved only declared encrypted vault references after authorization. Pinned public DNS, denied redirects and special ranges, scrubbed process environments, and required no-network Docker for Worker commands.
W6.4	Reserved and settled durable workspace/deployment rolling provider cost and billable actions atomically. Reconciled abandoned holds after lease expiry.
W6.5	Added one-shot executor permits and a production-catalog bypass matrix so direct handler access fails before secrets, budgets, effects, or I/O.

Security ordering



Fail-closed drivers

Worker browser, SMTP, and SQL paths remain disabled until hardened drivers reach the same boundary quality. Existing inherited Agent adapters do not silently become canonical Worker permissions.

W7: attention, approval, and independent control

Human intervention is durable state, not a transient UI prompt.

Durable graph

- Version-bound attention requests, approval grants, control commands, and legacy-compatible notification references.
- Digest-only approval nonces, exact operation and policy bindings, expiration, escalation, and replay identities.
- Atomic pause, resume, stop, deployment revoke, approve once, approve for run, and reject commands with expected revisions.

Supervisor integration

- Approval-required operations checkpoint the exact pending action before the run enters waiting state.
- Escalation and expiration jobs are bounded and deduplicated. Expiration pauses or rejects according to immutable policy.
- Final tool execution rechecks the grant at the boundary; a one-time grant can replay only its original action.

Independent operator surface

- The Worker operator API is mounted independently from authoring services and enforces per-action roles.
- Stop and revoke remain usable after fresh process reconstruction.
- Race tests cover approve versus reject, activation versus revoke, and stop at every supervisor phase.

Gate result

A person can stop a run or revoke future activation without PacketADE or any authoring application being present.

W8: evidence, rollups, retention, API, and UI

One server-side read answers who ran, why, under which immutable version, within which budgets, what evidence exists, and why the run stopped.

Slice	Answerability delivered
W8.1	Versioned monotonic Worker events paired atomically with digest-bound evidence. Artifact manifests bind content and provenance.
W8.2	Deterministic cumulative rollups by immutable version, deployment, and run. Missing sources become typed gaps instead of silent zeroes.
W8.3	Bounded redaction and retention by category, digest-checked artifact deletion, deletion evidence, active-run preservation, and dry runs.
W8.4	Workspace-scoped health/list/detail/cursor APIs, bounded resumable SSE, and accessible canonical Worker list/detail workbench.
W8.5	One-read answerability, source-order replay, secret-free reads, tenant-bound cursors, accessible states, and backend parity gate.

Retention invariant

Metadata may compact only when the surviving tombstone preserves enough digest-bound provenance to explain the resulting source gap. Active or resumable work is never compacted merely because it is old.

W9: PacketADE deployment handoff

PacketAgent now implements the receiving side of a strict, authenticated, resumable PacketADE handoff.

WorkerPackage v1

- Strict W1-aligned JSON schema with canonical UTF-8 bytes, mandatory SHA-256 package digest, no inline secrets, and rejection of undeclared fields or unknown major versions.
- Optional DSSE payload/signature verification seam bound to the same canonical bytes.

Trust and deployment

- Workspace, actor, operation, expiry, revocation, rate, and optional signature requirements are bound to PacketADE service credentials.
- Local capability acceptance narrows the package upper bound before durable integrity, provenance, policy, actor, and idempotency receipts are stored.
- Validate, deploy, update, activate, inspect, list runs, pause, resume, rollback, and revoke compose W2, W3, W7, and W8 services.

Reconnectable events

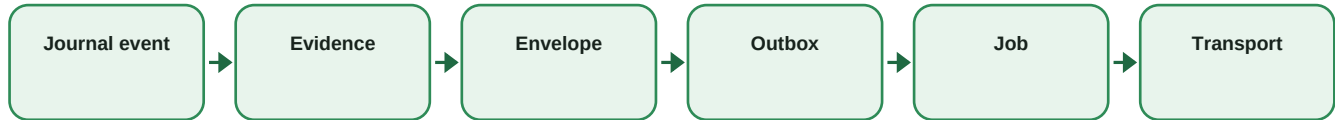
- Versioned deployment and run event pages plus bounded SSE with stable cursor IDs, evidence links, explicit trace gaps, and Last-Event-ID resume.
- Durable cursor acknowledgement is an explicit idempotent ETag-guarded write; opening or closing a stream never acknowledges delivery.
- The W9.5 gate serializes the store, reconstructs fresh services, reconnects evidence, then updates, pauses, resumes, rolls back, and revokes.

Scope truth

No PacketADE repository was modified in this workstream. PacketAgent's receiving contract and local fake-client interoperability are implemented; live PacketADE integration depends on the external product.

W10.1: channel-neutral notification outbox

The outbox separates durable Worker truth from any individual delivery channel.



- packetagent.worker-notification-envelope/v1 carries CloudEvents-inspired source, id, type, subject, and time plus immutable Worker/version/run, source event, evidence, and thread bindings.
- packetagent.worker-notification-outbox/v1 stores a stable delivery and idempotency key, route kind/reference, attempt lease, retry bounds, expiry, dead-letter state, and allowlisted delivery metadata.
- Attention, progress checkpoint, and terminal event/evidence/outbox/job records commit in one transaction.
- Pending deliveries pin source evidence. Delivered, dead-letter, or expired items can compact only behind a digest-bound tombstone.
- The scheduler can re-drive an abandoned attempt with the same idempotency key; unconfigured external transports fail closed.

Standards

The envelope follows the CloudEvents 1.0 core shape. Retry behavior relies on independent idempotency semantics consistent with RFC 9110 section 9.2.2.

W10.2: PacketChat delivery - message contract

W10.2 is implemented inside PacketAgent and committed as b259a4f.

Encrypted route configuration

- External routes must use a vault:* reference declared by the immutable Worker version.
- packetagent.packetchat-route/v1 keeps endpoint, optional bearer token, callback origin, callback secret, timeout, and callback lifetime encrypted at rest.
- Workspace admins use metadata-only GET, PUT, and DELETE /api/app/workers/credentials. No plaintext, ciphertext, IV, or authentication tag is returned.

Bounded Worker card

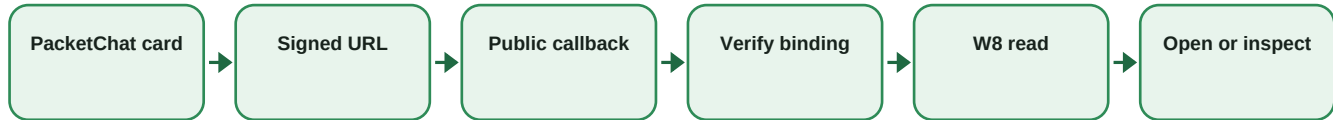
- packetagent.packetchat-worker-message/v1 includes exact workspace, definition, deployment, run, version ID, version number/status, and content digest.
- The card contains deployment/run status and reason, budget policy and usage, latest checkpoint, evidence link, required action, and bounded title/summary.
- All progress for one run uses one stable thread and replaceable progress message key. Attention and terminal cards append.
- The W10.1 idempotency key is sent in the Idempotency-Key header.

Hardened delivery

- The transport reloads the durable run, deployment, version digest, and route before resolving the credential.
- Outbound POST uses the W6 pinned-DNS client, redirect denial, bounded timeout, bounded response, and optional bearer header.
- 408, 425, 429, 5xx, and transient network errors retry within outbox bounds. Configuration failures and other 4xx outcomes terminate safely.

W10.2: PacketChat delivery - callbacks and security

Read-only callbacks are authenticated without weakening W7. Mutating remote control remains W10.3.



Compact HS256 JWT

- Header is fixed to alg HS256 and typ JWT. Claims bind issuer PacketAgent, audience PacketChat, action, subject, issued time, expiry, and JTI.
- Claims also bind exact workspace, definition, deployment, run, version ID, version content digest, route ID, and route reference.
- Signature comparison is constant-time. Weak callback secrets, non-origin callback bases, unsupported algorithms, future issue times, excessive lifetimes, tampering, expiry, and binding drift fail closed.
- Public callback failures use one generic external code to avoid exposing a Worker-existence oracle. Access logs redact the token query parameter.

Callback outcomes

- open returns only the matching /runs/worker/:runId workbench URL.
- inspect returns W8's redacted server-side Worker detail.
- Responses carry Cache-Control: no-store and Pragma: no-cache.
- Signed callback values are outbound-only and never written to store, event, evidence, outbox, job, or metadata.

Important boundary

W10.2 callbacks are intentionally read-only and replay-safe within their short lifetime. W10.3 must add durable nonce consumption before PacketPhone can mutate approvals or kill state.

Persistence, verification, and known boundaries

The gate is broad, but it is not a claim that every external Packet product has been changed or deployed.

Verification	Result
API suite	1,493 passed, 2 intentionally skipped, 0 failed.
Web suite	28 passed, 0 failed.
TypeScript	Backend and web typecheck passed.
Production web build	Vite build passed.
Lint	0 errors; 145 inherited warnings remain in R1.
Persistence	JSON, SQLite, and managed Postgres parity gates cover all persisted Worker record changes.
W10.2 focused	Encrypted vault round trip, payload bounds, idempotency, secret isolation, route validation, retry classes, signature tamper/expiry/workspace binding, redacted inspect, no-store route.
Live interoperability	Not run. No PacketChat or PacketPhone endpoint and credential pair was configured.

Public claim boundary	Do not describe the unified Packet-suite Worker lifecycle as shipped until W10.4 passes. PacketPhone controls do not exist yet, and PacketChat has only local contract certification.
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Pause handoff and remaining ledger

The exact resume point is W10.3 in BACKLOG.md. That file is the sole active remaining-work ledger.

Next: W10.3 PacketPhone controls

- Deliver approve, reject, pause, stop, and revoke only through W7 actor and role checks.
- Bind tokens to action, workspace, deployment, run, attention request, version digest, actor, audience, nonce, and expiry.
- Consume mutating callbacks durably and reject stale, replayed, cross-workspace, cross-version, and already-resolved actions.

Then: W10.4 remote-control certification

- Contract-test PacketChat and PacketPhone adapters against fake endpoints.
- Race local and remote actions, rotate credentials, restart pending deliveries, replay callbacks, and recover dead letters.
- Run live interoperability when external endpoints and credentials are available.

After W10

- R1 repository health and historical finding re-audit.
- R2 provider policy and key parity.
- R3 file-tree generation depth.
- R4 generated-app runtime and self-host publish.
- R5 sandbox, egress, and preview isolation.
- R6 agent authoring and execution depth.
- R7 Builder and frontend maintainability.
- R8 release reliability and production packaging.

Authority

dev/CODEX-HANDOFF.md points to BACKLOG.md. dev/worker-implementation-loops.md explains execution mechanics only and cannot introduce active work absent from the backlog.

Implementation commit ledger and references

Each major loop was committed as an isolated, reviewable slice.

Commit	Slice
30bdc17	W1 canonical Worker contract
640fcda	W2 durable lifecycle
61a763d	W3 activation intake
768dda2	W4 bounded supervisor
f493496	W5 recovery and effect safety
1191a9f - 42202d3	W6 permissions, boundaries, budgets, bypass gate
359ce10 - 5ea1330	W7 controls, attention, operator API, restart/kill gate
6d1797c - 8e6add0	W8 evidence, rollups, retention, API/UI, answerability gate
9cd96de - 17191e4	W9 WorkerPackage, trust, deployment, events, restart handoff
55a85c7	W10.1 notification outbox
b259a4f	W10.2 PacketChat Worker delivery

Repository: [git@github.com:packetloss404/PacketAgent.git](https://github.com:packetloss404/PacketAgent.git)

Primary branch: main

Standards references: CloudEvents 1.0 core specification; RFC 9110 section 9.2.2; RFC 7515 JSON Web Signature; RFC 7519 JSON Web Token.

Canonical sources: README.md, BACKLOG.md, dev/CODEX-HANDOFF.md, dev/worker-implementation-loops.md, dev/TESTING.md, and CHANGELOG.md.